Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/02/2026
PI Name:	Yamamoto, Namiko Ph.D.		
Project Title:	Enhancing Our Understanding of Cement Hydration in a Micro-Gravity Setting and its Mechanical Behaviors Through the Application of Machine Learning-Aided 3D Reconstruction and Characterization of Micro-Scale Structures		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-1401	**Congressional District:**	12
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2023 NNH23ZDA001N-PSI E.8 Physical Sciences Informatics
Start Date:	10/01/2024	**End Date:**	09/30/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: Start date changed to 10/01/2024 per the PI (Ed., 7/2/26).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Radlinska, Aleksandra Ph.D. (Pennsylvania State University) Saseendran, Vishnu Ph.D. (Pennsylvania State University) Thomas, Robert Ph.D. (Clarkson University)		
Grant/Contract No.:	80NSSC25K7100		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of the proposed work is to assess the effectiveness of nanoseed addition during cement hydration in the microgravity environment for enhanced mechanical performance. Concrete-like materials will be essential in the development of infrastructure on the Moon and Mars. The study of cement hydration behaviors in the microgravity environments has been initiated by Co-Principal Investigator (Co-PI) Radlińska with the NASA-funded projects titled as, "Microgravity Investigation of Cement Solidification (MICS)" and as "MICS-MVP (Multi-use Variable-gravity Platform) Microgravity Investigation of Cement Solidification", and then the extension of this work using the Multi-use Variable-gravity Platform, where various cements, including lunar regolith simulants, were hydrated on the International Space Station. Some of these samples have been characterized using nanoindentation by Co-PI Thomas via NASA Cooperative Agreement Notices (CAN) funding. Yet currently, two major challenges exist about space-originated samples. First, cements hydrated under the microgravity exhibited larger porosity than when hydrated on ground, as the particles do not sediment. Here, nanoseed addition is expected to enhance nucleation, rapidly form phases, and mitigate the porosity, which is advantageous to improve the cements' mechanical properties; the NASA Physical Sciences Informatics (PSI) database offers 2D scanning electron microscopy images of C3S cement samples hydrated with nanoseeds. Second, the small size of existing space-hydrated cement samples is limiting their mechanical characterization at the macro scale. With the sample sizes on the order of few centimeters, and with the relatively large individual phase size on the order of 100s of microns, their effective mechanical properties cannot be characterized; nanoindentation provides mechanical properties of individual phases. Recently, Co-PIs Saseendran and Yamamoto developed a framework to model mechanical behaviors using the 2D backscattered electron images of a simple cement sample available from the NASA PSI database. In this proposed work, first, the 2D images will be analyzed to capture the micro-structures, including porosity and its distribution, of the cement samples hydrated in space with nanoseeds quantitatively, to complete the process-structure relationship study. Second, AI-assisted codes will be used to reconstruct 3D microstructures from the available 2D images, and such 3D microstructure volumes will be inputted into a micromechanics-based tool to evaluate their mechanical properties. Two deliverables from this project will include: 1) the knowledge about nanoseed effects on micro-structures and mechanical properties; and 2) the framework to analyze and model the micro-structures; and they can be applied to other relevant materials including cement-like mixtures from lunar or Mars regolith in future, accelerating the process to identify the cement mixture and hydration process, per the gravity environment, to achieve the desired mechanical properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The two major deliverables from this proposed work are: 1) the knowledge on nanoadditive effects on micro-structures and associated mechanical behaviors of C3S cement hydrated in the microgravity environment; and 2) the framework to characterize the micro-structures and to model mechanical behaviors, including damage, from 2D scanning electron microscopy (SEM) images through machine learning (ML). Going through the larger space-hydrated samples of varying microstructures, the first training data set will be provided for ML-assisted 3D reconstruction of micro-structures only from 2D images, eventually making 3D microCT scans possible for future samples. Both deliverables can be applied to other relevant materials, including cement-like mixtures from lunar or Mars regolith in future. The above developed tool to acquire the process-structure-property relationship and the obtained relationship knowledge are expected to accelerate the process to identify the cement mixture and hydration process, per the gravity environment, to achieve the desired mechanical properties.
Task Progress:	August 2025 Update per the Principal Investigator (PI): The goal is to assess the effectiveness of nanoseed addition during cement hydration in the microgravity environment for enhanced mechanical performance. Nanoseed addition is expected to enhance nucleation, rapidly form phases, and mitigate the porosity increase due to decreased gravity observed in the previous study. As part of the Microgravity Investigation of Cement Solidification (MICS) project, cement samples (C3S phase) were hydrated on the International Space Station (ISS); their scanning electron microscope (SEM) images are available on the NASA Physical Sciences Informatics (PSI) database, and the samples are currently housed in PI Radlińska's lab as more analysis becomes necessary. (Ed. note: FY25 update added 7/2/26. Project added to Task Book 2/2/26).
Bibliography Type:	Description: (Last Updated: 09/26/2024)